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LOGICAL QUBIT ENCODING SURFACE

Abstract

A quantum computing device is provided, including a logical qubit encoding surface including a plurality of plaquettes. Each plaquette of the plurality of plaquettes may include a plurality of measurement-based qubits. The plurality of measurement-based qubits may include four data qubits and a first ancilla qubit. The first ancilla qubit may be electrically connected to the four data qubits and a second ancilla qubit included in the logical qubit encoding surface.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/174,616, filed Feb. 25, 2023, which is a continuation of U.S. patent application Ser. No. 16/657,922, filed Oct. 18, 2019, now U.S. Pat. No. 11,599,817, the entirety of each of which is hereby incorporated herein by reference for all purposes.

BACKGROUND

[0002] Errors frequently occur when performing computations with a quantum computing device. In order for the results of such computations to be accurate, the errors typically have to be corrected at a faster rate than the rate at which they occur.

[0003] The error rate for quantum computations depends on structural properties of the quantum computing device. For example, increasing the number of physical qubits used to implement a logical qubit may increase the error rate. The error rate of a physical qubit may also increase as the connectivity degree of the physical qubit (the number of other physical qubits to which the physical qubit is connected) increases.

[0004] The architecture of a quantum computing device also affects what error correction methods may be used. Accordingly, it is beneficial for a quantum computing device to have a structure that both has a low error rate and allows efficient methods of error correction to be used.

SUMMARY

[0005] According to one aspect of the present disclosure, a quantum computing device is provided, including a logical qubit encoding surface including a plurality of plaquettes. Each plaquette of the plurality of plaquettes may include a plurality of measurement-based qubits. The plurality of measurement-based qubits may include four data qubits and a first ancilla qubit. The first ancilla qubit may be electrically connected to the four data qubits and a second ancilla qubit included in the logical qubit encoding surface.

[0006] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIGS. 1A-1B show an example logical qubit encoding surface including a plurality of CNOT-based qubits, including a 5×5 grid of data qubits.

[0008] FIG. 2 shows an example logical qubit encoding surface including a plurality of measurement-based qubits, including a 5×5 grid of data qubits.

[0009] FIG. 3 shows an example logical qubit encoding surface including a plurality of measurement-based qubits, including a 3×3 grid of data qubits.

[0010] FIG. 4 shows an example logical qubit encoding surface including a plurality of measurement-based qubits, including a 7×7 grid of data qubits.

[0011] FIG. 5 schematically shows an example quantum computing device, according to the

embodiment of FIG. 2.

[0012] FIG. 6 shows an example measurement sequence with which a controlled not (CNOT) gate may be implemented, according to the embodiment of FIG. 2.

[0013] FIG. 7 shows a stabilizer operator measurement sequence that may be used to measure an XXXX stabilizer operator, according to the embodiment of FIG. 2.

[0014] FIG. 8 shows a measurement sequence with which a two-target CNOT gate may be implemented, according to the embodiment of FIG. 2.

[0015] FIG. 9 shows another stabilizer operator measurement sequence that may be used to measure an XXXX stabilizer operator, according to the embodiment of FIG. 2.

[0016] FIG. 10 shows a generalization of the stabilizer operator measurement sequence of FIG. 9.

[0017] FIG. 11 shows an example measurement sequence that may be used to measure a Pauli XXX operator, according to the embodiment of FIG. 2.

[0018] FIG. 12 shows an example measurement sequence with which a plurality of two-qubit CNOT gates may be measured, according to the embodiment of FIG. 2.

[0019] FIG. 13 shows another example configuration of a logical qubit encoding surface including a plurality of measurement-based qubits, including a 5×5 grid of data qubits.

[0020] FIG. 14 shows another example configuration of a logical qubit encoding surface including a plurality of measurement-based qubits, including a 5×5 grid of data qubits.

[0021] FIG. 15A shows a flowchart of an example method for performing a measurement of a logical qubit with a quantum computing device.

[0022] FIG. 15B shows additional steps of the method of FIG. 15A that may be performed when measuring a stabilizer operator.

[0023] FIG. 16 shows a flowchart of an example method for performing a surface code error correction at a quantum computing device.

[0024] FIG. 17 shows a schematic view of an example computing environment in which the quantum computing device of FIG. 5 may be enacted.

DETAILED DESCRIPTION

[0025] Surface code error correction is one existing method of error correction that has been developed for use in quantum computing devices. Surface code error correction may be used in quantum computing devices that include physical qubits that have controlled not (CNOT) gates and are arranged in a substantially planar layout. Surface codes may allow error correction to be performed for physical qubits that have high error rates. In addition, surface codes may be implemented entirely with single-qubit gates and nearest-neighbor operations.

[0026] FIG. 1A shows an example conventional logical qubit encoding surface 10 at which a surface code may be used. Each physical qubit included in the logical qubit encoding surface 10 of FIG. 1A is a CNOT-based qubit. The logical qubit encoding surface 10 of FIG. 1A includes a plurality of plaquettes 12 that each include four data qubits 14. Each data qubit 14 may be electrically connected to its vertical and horizontal neighbors to form a rectangular grid. As shown in FIG. 1A, the plurality of data qubits 14 included in the logical qubit encoding surface 10 may form a square grid. In addition, the data qubits 14 included in each plaquette 12 may be arranged in a square shape.

[0027] Each plaquette 12 may further include an ancilla qubit 16, which may be electrically coupled to each of the four data qubits 14 included in the plaquette 12. Each of the electrical connections 18 between a data qubit 14 and an ancilla qubit 16 shown in FIG. 1A includes a CNOT gate 20. In addition, a plurality of ancilla qubits 16 may be located outside the edges of the rectangular grid formed by the plurality of data qubits 14. Such ancilla qubits 16 may be electrically connected to pairs of data qubits 14 such that each data qubit 14 located at an edge of the rectangular grid is connected to one ancilla qubit 16 located outside the edges of the rectangular grid. FIG. 1B shows the conventional logical qubit encoding surface 10 of FIG. 1A including the plurality of ancilla qubits 16.

[0028] When surface code error correction is performed, stabilizer operators XXXX and ZZZZ may be measured at the ancilla qubits **16** of the logical qubit encoding surface **10**. The XXXX stabilizer operator includes an Pauli X operator applied to each of the data qubits **14** of a plaquette **12**, and the ZZZZ stabilizer operator includes a Pauli Z operator applied to each of the data qubits **14** of a plaquette **12**. The XXXX stabilizer operator and the ZZZZ stabilizer operator may be measured at ancilla qubits **16** having alternating positions in the rectangular grid. FIG. **1B** shows a plurality of XXXX ancilla qubits **16A** that alternate with a plurality of ZZZZ ancilla qubits **16B**. Thus, the logical qubit encoding surface **10** may include a plurality of XXXX plaquettes **12A**, each of which includes an XXXX ancilla qubit **16A**, and a plurality of ZZZZ plaquettes **12B**, each of which includes a ZZZZ ancilla qubit **16B**. In addition, XX stabilizer operators and ZZ stabilizer operators may be respectively measured at the ancilla qubits **16A** and **16B** that are located outside the edges of the rectangular grid and are each connected to only two data qubits **14**.

[0029] The respective measurement of each stabilizer operator may output one bit, known as a syndrome bit. Thus, a plurality of syndrome bits may be measured for different locations on the logical qubit encoding surface. The plurality of syndrome bits may be conveyed to a classical computing device, at which a decoder may determine, based on the plurality of syndrome bits, a location on the logical qubit encoding surface at which an error has occurred.

[0030] In the example of FIGS. **1A-1B**, for each data qubit **14** included in a plaquette **12**, a respective CNOT gate **20** is applied to that data qubit **14** along the electrical connection **18** between that data qubit **14** and the ancilla qubit **16** of the plaquette **12**. The CNOT gate **20** entangles the data qubit **14** and the ancilla qubit **16** such that the measurement performed on the ancilla qubit **16** is a joint measurement of the data qubit **14** and the ancilla qubit **16**.

[0031] Measurement-based qubits are physical qubits for which the available measurements are one-qubit and two-qubit measurements of the X, Y, and Z Pauli operators. A measurement-based qubit may, for example, be constructed from a plurality of Majorana zero modes (MZMs). A Pauli operator of the measurement-based qubit may be measured by measuring a joint fermionic parity of a proper set of the plurality of the MZMs. Such measurements of Pauli operators may be topologically protected, thereby reducing the error rate of the measurements.

[0032] Although measurement-based qubits do not have native CNOT gates, a CNOT gate may be constructed from a plurality of Pauli measurements, as discussed in further detail below. However, if measurement-based qubits were used in place of the CNOT-based data qubits **14** in the configuration shown in FIGS. **1A-1B**, the resulting logical qubit encoding surface **10** would include too few ancilla qubits **16** for surface code error correction to be used.

[0033] In order to form CNOT gates **20** when measurement-based qubits are used, an additional ancilla qubit **16** may be added along each of the electrical connections **18** of the logical qubit encoding surface **10** shown in FIG. **1B**. However, such a configuration would include **104** ancilla qubits **16**, whereas the configuration of FIG. **1B** includes only **24** ancilla qubits **16**. This increase in the number of ancilla qubits **16** would significantly increase the error rate and the cost of manufacturing the logical qubit encoding surface **10**.

[0034] To address these issues associated with the conventional logical qubit encoding surface **10** of FIGS. **1A** and **1B**, as an alternative to a configuration in which an additional ancilla qubit **16** is included for each CNOT gate **20**, a logical qubit encoding surface **110** according to the present disclosure is shown in FIG. **2**. The logical qubit encoding surface **110** of FIG. **2** includes a plurality of plaquettes **112**. Each plaquette **112** of the plurality of plaquettes **112** includes a plurality of measurement-based qubits, including four data qubits **114** and a first ancilla qubit **116**. The first ancilla qubit **116** may be electrically connected to the four data qubits **114**. In contrast to the electrical connections **18** shown in FIGS. **1A-1B**, the electrical connections **118** shown in FIG. **2** do not include native CNOT gates **20**.

[0035] In some embodiments, as shown in the example of FIG. **2**, the four data qubits **114** of the

plaquette **112** may be arranged in a square. In such embodiments, the first ancilla qubit **116** may be located within the square. The respective data qubits **114** included in the plurality of plaquettes **112** may also be arranged in a rectangular grid. In the example of FIG. 2, the rectangular grid is a 5×5 square grid.

[0036] The first ancilla qubit **116** may be electrically connected to a second ancilla qubit **126**. The second ancilla qubit **126** may be included in the logical qubit encoding surface **110** outside the plaquette **112** in which the first ancilla qubit **116** is located. The second ancilla qubit **126** may also be electrically connected to two of the four data qubits **114** of the plaquette **112**. For example, the second ancilla qubit **126** may be included in a neighboring plaquette **112**. Alternatively, in embodiments in which the plurality of data qubits **114** form a rectangular grid, the second ancilla qubit **126** may be located outside an edge of the rectangular grid.

[0037] In embodiments in which the plurality of data qubits **114** form a rectangular grid, the electrical connections **118** between first ancilla qubits **116** and second ancilla qubits **126** may form a “windmill” pattern centered around the middle of the rectangular grid. In such embodiments, for each plaquette **112** of the plurality of plaquettes **112**, the electrical connection **118** between the first ancilla qubit **116** and the second ancilla qubit **126** may extend in a direction perpendicular to the edge of the rectangular grid that is closest to a midpoint between the first ancilla qubit **116** and the second ancilla qubit **126**.

[0038] The logical qubit encoding surface **110** of FIG. 2 includes a total of **24** ancilla qubits. The configuration of FIG. 2 therefore allows measurement-based qubits to be used in place of CNOT-based qubits without increasing the total number of ancilla qubits included in the logical qubit encoding surface **110** relative to the logical qubit encoding surface **10** of FIGS. 1A-1B.

[0039] FIG. 3 shows another example logical qubit encoding surface **210** that includes a 3×3 square grid of data qubits **114**. In the example of FIG. 3, the logical qubit encoding surface **210** includes four plaquettes **112** and eight ancilla qubits. In addition, FIG. 4 shows another example logical qubit encoding surface **310** that includes a 7×7 square grid of data qubits **114**. The logical qubit encoding surface **310** of FIG. 4 includes **36** plaquettes **112** and **48** ancilla qubits. The logical qubit encoding surface **210** of FIG. 3 and the logical qubit encoding surface **310** of FIG. 4 each include the same total number of ancilla qubits as would be included in 3×3 and 7×7 versions of the CNOT-based logical qubit encoding surface **10** of FIGS. 1A-1B.

[0040] Turning now to FIG. 5, a quantum computing device **100** is schematically depicted. As shown in FIG. 5, the quantum computing device **100** may include one or more logical qubit encoding surfaces **110** with which quantum computations may be performed. Although the quantum computing device **100** depicted in FIG. 5 includes one or more copies of the logical qubit encoding surface **110** of FIG. 2, the quantum computing device **100** may alternatively include one or more copies of the logical qubit encoding surface **210** of FIG. 3, the logical qubit encoding surface **310** of FIG. 4, or a logical qubit encoding surface with some other configuration.

[0041] The quantum computing device **100** of FIG. 5 may further include a measurement device **130**. The measurement device **130** may, for example, be a circuit that is electrically connected to the logical qubit encoding surface **110**. For each measurement qubit encoding surface **110**, the measurement device **130** may be configured to perform a measurement of a logical qubit encoded by that logical qubit encoding surface **110**. In addition, the measurement device **130** may be further configured to measure a respective stabilizer operator of each plaquette **112** included in the logical qubit encoding surface **110**, as discussed in further detail below. Each measurement of the stabilizer operator of a plaquette **112** may result in one bit, known as a syndrome bit **132**.

[0042] The quantum computing device **100** may further include a classical computing device **140**, as shown in the example of FIG. 5. The classical computing device **140** may include a processor **142** and memory **144**, which may be communicatively coupled. In some embodiments, the classical computing device may further include one or more input devices, one or more output devices, and/or one or more communication devices. In some embodiments, the functions of the

classical computing device **140** may be distributed across a plurality of physical computing devices that are communicatively coupled via their respective communication devices.

[0043] The classical computing device **140** may be configured to receive the one or more syndrome bits **132** from the measurement device **130**. The one or more syndrome bits **132** may be received at a decoder **150** executed by the processor **142** of the classical computing device **140**. At the decoder **150**, the processor **142** may be configured to identify an error **152** in the measurement of the logical qubit based on the one or more syndrome bits **132** obtained as respective measurement results of the stabilizer operators. For example, the decoder **150** may identify the error **152** based on the one or more syndrome bits **132** using the union-find decoding algorithm. Since the union-find decoding algorithm is almost linear in complexity as a function of the number of physical qubits, the union-find algorithm may allow the classical computing device **140** to efficiently identify the error **152**.

[0044] As discussed above, for each plaquette **112** for which the stabilizer operator is measured, the stabilizer operator may be a product of four Pauli X operators or four Pauli Z operators of the four respective data qubits **114** included in the plaquette **112**. When measuring the stabilizer operators of the plaquettes **112** included in the logical qubit encoding surface **110**, the measurement device **130** may be configured to alternate between measuring the XXXX stabilizer operator and the ZZZZ stabilizer operator. Returning to FIG. 2, the measurement device **130** may be configured to measure the product of the four Pauli X operators for each of a first set of plaquettes **112A** and the product of the four Pauli Z operators for each of a second set of plaquettes **112B**. In the example of FIG. 2, each plaquette **112A** included in the first set of plaquettes **112A** includes an XXXX ancilla qubit **116A**. In addition, each plaquette **112B** included in the second set of plaquettes **112B** includes a ZZZZ ancilla qubit **116B**.

[0045] Measuring a stabilizer operator at the logical qubit encoding surface **110** may include applying a CNOT gate to at least one ancilla qubit. Although measurement-based qubits do not have native CNOT operations, a CNOT gate may be constructed using a sequence of Pauli operator measurements. Turning now to FIG. 6, a measurement sequence **160** with which a one-target CNOT gate **120** may be implemented using measurement-based qubits is shown. The measurement sequence **160** of FIG. 6 is performed with a first qubit **1** that is used as a control qubit, a second qubit **2** that is used as an ancilla qubit, and a third qubit **3** that is used as a target qubit. When an XXXX stabilizer operator is measured for a plaquette **112**, the two ancilla qubits used in the measurement may correspond to the first qubit **1** and the second qubit **2** shown in FIG. 6, and when a ZZZZ stabilizer operator is measured, the two ancilla qubits used in the measurement may correspond to the second qubit **2** and the third qubit **3**.

[0046] In some embodiments, the third qubit **3** may be the second ancilla qubit **126** of the plaquette **112**. In such embodiments, applying the CNOT gate **120** may include performing a plurality of first ancilla measurements of at least the first ancilla qubit **116**. Each first ancilla measurement may be a one-qubit measurement or a two-qubit measurement.

[0047] The measurement sequence **160** shown in FIG. 6 includes a plurality of measurements of Pauli X operators and Pauli Z operators, each of which are depicted as squares in FIG. 6. Each measurement may be a single-qubit measurement or a two-qubit measurement. The measurement sequence **160** shown in FIG. 6 also includes a plurality of Pauli updates, which are shown as rounded squares. Each Pauli update may apply a Pauli operator to a qubit or a pair of qubits under the condition that the preceding measurement had a nontrivial outcome. A nontrivial outcome is defined in the example of FIG. 6 as a measurement outcome of $|-\rangle$, which is returned when an error **152** has occurred. In the example of FIG. 6, each Pauli update applies a Pauli X operator or a Pauli Z operator. In embodiments in which the third qubit **3** is the second ancilla qubit **126**, implementing the CNOT gate **120** may include, for each first ancilla measurement, performing a Pauli update on at least the first ancilla qubit **126** subsequently to the first ancilla measurement.

[0048] FIG. 7 shows a stabilizer operator measurement sequence **162** that may be used to measure an XXXX stabilizer operator at a plaquette **112**. The physical qubits of the plaquette **112** are

labeled as a first qubit **1**, a second qubit **2**, a third qubit **3**, a fourth qubit **4**, a fifth qubit **5**, and a sixth qubit **6**. The fifth qubit **5** and the sixth qubit **6** are a first ancilla qubit **116** and a control qubit respectively, and the other four qubits are data qubits **114**. The measurement sequence **162** of FIG. 7 includes sixteen measurement timesteps in which a measurement and a corresponding Pauli update are performed.

[0049] When the measurement device **130** measures the stabilizer operator for the plaquette **112**, the measurement device **130** may be configured to prepare the second ancilla qubit **126** (the sixth qubit **6** in the example of FIG. 7) to have a $|+\rangle$ state prior to performing the measurement. Alternatively, the measurement device may prepare the second ancilla qubit **126** to have a $|-\rangle$ state. The measurement device **130** may be further configured to apply one or more respective CNOT gates **120** to the plaquette **112**. The first ancilla qubit **116** may be used as the ancilla qubit of the CNOT gate **120** and may correspond to the fifth qubit **5** shown in FIG. 7. In addition, a data qubit **114** of the four data qubits **114** may be the target qubit of the CNOT gate **120**. The second ancilla qubit **126** of the plaquette **112** may be used as the control qubit. In the example of FIG. 7, measuring the XXXX stabilizer operator includes applying four CNOT gates **120**, each of which has a different one of the four data qubits **114** as the target qubit.

[0050] Subsequently to applying the one or more CNOT gates **120** to the plaquette **112**, measuring the stabilizer operator may further include performing a second ancilla measurement at the second ancilla qubit **126**. The second ancilla measurement performed at the second ancilla qubit **126** may be a measurement of a Pauli X operator when the XXXX stabilizer operator is measured or a Pauli Z operator when the ZZZZ stabilizer operator is measured. The final measurement result is obtained from the measurement of the sixth qubit **6**, as indicated with a vertical line in the example of FIG. 7.

[0051] FIG. 8 shows a measurement sequence **164** with which a two-target CNOT gate **122** may be implemented. In the example of FIG. 8, the first qubit **1** and the second qubit **2** are target qubits, the third qubit **3** is an ancilla qubit, and the fourth qubit **4** is the control qubit.

[0052] FIG. 9 shows another stabilizer operator measurement sequence **166** by which the measurement device **130** may be configured to measure the XXXX stabilizer operator. In the example of FIG. 9, CNOT gates **120** are implemented by applying two two-target CNOT gates **122** to the plaquette **112**. The first two-target CNOT gate **122** included in the measurement sequence **166** of FIG. 9 has the first qubit **1** and the second qubit **2** as its target qubits. The second two-target CNOT gate **122** included in the measurement sequence **166** has the third qubit **3** and the fourth qubit **4** as its target qubits. The stabilizer operator measurement sequence **164** of FIG. 9 includes ten measurement timesteps rather than the sixteen measurement timesteps included in the measurement sequence **162** of FIG. 7. Thus, the measurement sequence **166** of FIG. 9 allows the XXXX stabilizer operator to be measured more quickly and with a lower error rate than the measurement sequence **162** of FIG. 7.

[0053] FIG. 10 shows a stabilizer operator measurement sequence **168** that generalizes the XXXX stabilizer operator measurement sequence of FIG. 9 to plaquette architectures including any even number $n \geq 2$ of data qubits **114**. In the measurement sequence **168** of FIG. 10, a two-target CNOT gate **122** is performed for each pair of target qubits that are adjacent in the numbering order such that each data qubit **114** is used as a target qubit in one two-target CNOT gate **122**. The first qubit **1** and the second qubit **2** are the first ancilla qubit and the second ancilla qubit in the example of FIG. 10.

[0054] FIG. 11 shows an example measurement sequence **170** that may be used to measure a Pauli XXX operator **124**. The Pauli XXX operator **124** may be used, for example, when measuring a stabilizer operator in the Pauli X basis for a plaquette that has an odd number $n \geq 3$ of data qubits **114**, as shown in the example of FIG. 12. In the measurement sequence **172** shown in FIG. 12, a plurality of two-target CNOT gates **122** are measured, as in the measurement sequence **168** of FIG. 10. The first qubit **1** and the second qubit **2** are the first ancilla qubit **116** and the second ancilla

qubit **126** in the example of FIG. **12**. After measuring the two-target CNOT gates **122**, the measurement device **130** may be further configured to measure the Pauli XXX operator **124** for qubits n , $n+1$, and $n+2$.

[0055] Thus, the measurement device **130** may measure respective stabilizer operators for the plaquettes **112** of the logical qubit encoding surface **110** in order to perform error detection at the decoder **150**. These stabilizer operators may be measured for plaquettes including any number of measurement-based data qubits **114** greater than or equal to two. In addition, although FIGS. **6-12** show measurement sequences for measuring Pauli X operators, the measurement sequences of FIGS. **6-12** may be modified to measure Pauli Z operators instead. A measurement sequence in which a Pauli X operator is measured may be adapted to measure a Pauli Z operator by switching all Pauli X operators and Pauli Z operators in the measurement sequence, as well as switching all Pauli X updates and Pauli Z updates.

[0056] FIG. **13** shows an alternative configuration of a logical qubit encoding surface **410**, according to one example embodiment. The logical qubit encoding surface **410** of FIG. **13** includes a plurality of plaquettes **412**. In the example of FIG. **13**, the logical qubit encoding surface **410** includes sixteen plaquettes **412**. Each plaquette **412** of the plurality of plaquettes **412** includes a plurality of measurement-based qubits. The plurality of measurement-based qubits in each plaquette **412** includes four data qubits **114**, a first ancilla qubit **416**, and a second ancilla qubit **426**. In FIG. **13**, similarly to in FIG. **2**, the respective data qubits **114** included in the plurality of plaquettes **412** are arranged in a 5×5 rectangular grid. In addition, the four data qubits **114** of each plaquette **412** are arranged in a square.

[0057] In each plaquette **412** of the logical qubit encoding surface **410** of FIG.

[0058] **13**, the first ancilla qubit **416** is electrically connected to the four data qubits **114** and the second ancilla qubit **426**. However, each second ancilla qubit **426** is not electrically connected to any measurement-based qubits other than the first ancilla qubit **416**. Since each ancilla qubit included in the logical qubit encoding surface **410** of FIG. **13** is only used by one plaquette **412**, the total number of ancilla qubits increases to 48 compared to the 24 ancilla qubits included in the logical qubit encoding surface **110** of FIG. **2**. Although the total number of ancilla qubits is higher in the embodiment of FIG. **13** than in the embodiment of FIG. **2**, the connectivity degree of each qubit (the total number of other qubits to which that qubit is electrically connected) is at most four rather than five. This decrease in the maximum connectivity degree may reduce the amount of time elapsed per measurement cycle by a factor of two.

[0059] The logical qubit encoding surface **410** of FIG. **13** may be included in a quantum computing device **100**, which may include a measurement device **130** and a classical computing device **140**. Similarly to the logical qubit encoding surface **110** of FIG. **2**, the logical qubit encoding surface **410** of FIG. **13** alternates between a first set of plaquettes **412A** for which the measurement device **130** is configured to measure an XXXX stabilizer operator and a second set of plaquettes **412B** for which the measurement device **130** is configured to measure a ZZZZ stabilizer operator. Each plaquette **412A** of the first set of plaquettes **412A** may include a first XXXX ancilla qubit **416A** and a second XXXX ancilla qubit **426A**. In addition, each plaquette **412B** of the second set of plaquettes **412B** may include a first ZZZZ ancilla qubit **416B** and a second ZZZZ ancilla qubit **426B**.

[0060] A logical qubit encoding surface **510** is shown in FIG. **14** according to another example embodiment. In the embodiment of FIG. **14**, each plaquette **512** includes four data qubits **514** arranged in a square. Each plaquette **512** further includes a first ancilla qubit **516**, a second ancilla qubit **526**, and a third ancilla qubit **536**. The first ancilla qubit **516** and the second ancilla qubit **526** are each electrically connected to two of the data qubits **514** and to the third ancilla qubit **536**. Each data qubit **514** and each third ancilla qubit **536** of each plaquette **512** shown in FIG. **14** is connected to two first ancilla qubits **516** and two second ancilla qubits **526**.

[0061] The logical qubit encoding surface **510** of FIG. **14** may be included in a quantum computing

device **100**, which may include a measurement device **130** and a classical computing device **140**. In addition, the logical qubit encoding surface **510** may include a first set of plaquettes **512A** at which the measurement device **130** may be configured to measure an XXXX stabilizer operator and a second set of plaquettes **512B** at which the measurement device **130** may be configured to measure a ZZZZ stabilizer operator. Each plaquette **512A** of the first set of plaquettes **512A** may include a first XXXX ancilla qubit **516A** and a second XXXX ancilla qubit **526A**. In addition, each plaquette **512B** of the second set of plaquettes **512B** may include a first ZZZZ ancilla qubit **516B** and a second ZZZZ ancilla qubit **526B**.

[0062] FIG. **15A** shows a flowchart of an example method **600** for use with a quantum computing device. For example, the method **600** may be used with the quantum computing device **100** of FIG. **5** or with a quantum computing device having some other configuration. At step **602**, the method **600** may include performing a measurement of a logical qubit encoded by a logical qubit encoding surface including a plurality of plaquettes. The measurement may be performed using a measurement device included in the quantum computing device. Each plaquette of the plurality of plaquettes may include a plurality of measurement-based qubits. In some embodiments, the plurality of measurement-based qubits included in each plaquette includes four data qubits and a first ancilla qubit. The data qubits included in the logical qubit encoding surface may be arranged in a rectangular grid. The four data qubits may be arranged in a square, and the first ancilla qubit may be located within the square. In such embodiments, the first ancilla qubit may be electrically connected to the four data qubits and a second ancilla qubit included in the logical qubit encoding surface. The second ancilla qubit may be located inside or outside the plaquette.

[0063] At step **604**, the method **600** may further include measuring a respective stabilizer operator of each plaquette included in the logical qubit encoding surface. Each stabilizer operator may include a product of four Pauli X operators or four Pauli Z operators.

[0064] At step **606**, the method **600** may further include identifying an error in the measurement of the logical qubit based on respective measurement results of the stabilizer operators. The error in the measurement may be identified at a classical computing device communicatively coupled to the measurement device. A decoding algorithm such as the union-find decoding algorithm may be executed at the classical computing device to detect the error.

[0065] FIG. **15B** shows additional steps of the method **600** that may be performed in some embodiments when measuring the stabilizer operator for a plaquette. At step **608**, the method **600** may further include preparing the second ancilla qubit to have a $|+\rangle$ state. In other embodiments, the second ancilla qubit may be prepared to have a $|-\rangle$ state. At step **610**, the method **600** may further include applying one or more respective CNOT gates to the plaquette. For each CNOT gate, a data qubit of the four data qubits may be a target qubit. The first ancilla qubit of the plaquette may be the ancilla qubit of the CNOT gate. The second ancilla qubit connected to the plaquette may be the control qubit of the CNOT gate.

[0066] Applying the CNOT gate may include, at step **612**, performing a plurality of first ancilla measurements of at least the first ancilla qubit. Each first ancilla measurement may be a one-qubit measurement or a two-qubit measurement. At step **614**, applying the CNOT gate may further include, for each first ancilla measurement, performing a Pauli update on at least the first ancilla qubit. The Pauli update may include applying a Pauli X operator or a Pauli Z operator to the first ancilla qubit. Each Pauli update may be performed subsequently to its corresponding first ancilla measurement. In some embodiments, the Pauli update following a first ancilla measurement may be performed under the condition that the first ancilla measurement has a measurement outcome of $|-\rangle$.

[0067] The method **600** may further include, at step **616**, performing a second ancilla measurement at the second ancilla qubit subsequently to applying the one or more CNOT gates. The second ancilla qubit may be used as a control qubit in such embodiments. In some embodiments, the second ancilla measurement may be a measurement of a Pauli X operator or a Pauli Z operator. The

second ancilla measurement may return the value of the stabilizer operator for the plaquette.

[0068] In some embodiments, the method **600** may further include, at step **618**, measuring a product of four Pauli X operators for each of a first set of plaquettes and a product of four Pauli Z operators for each of a second set of plaquettes included in the logical qubit encoding surface. In such embodiments, the plurality of plaquettes may be arranged in a rectangular grid. The rectangular grid may include a first plurality of respective grid positions of the first set of plaquettes that alternate with a second plurality of respective grid positions of the second set of plaquettes.

[0069] FIG. **16** shows a flowchart of another example method **700** that may be performed with a quantum computing device. The quantum computing device at which the method **700** is performed may be the quantum computing device **100** of FIG. **5**. The method **700** may be performed additionally or alternatively to the method **600** of FIG. **15A**. At step **702**, the method **700** may include performing a surface code error correction at a logical qubit encoding surface including a plurality of measurement-based physical qubits. In embodiments in which the method **700** is performed in addition to the method **600** of FIG. **15A**, the surface code error correction may be performed subsequently to identifying an error in the measurement of the logical qubit at step **606**. [0070] At step **704**, the method **700** may include implementing one or more two-qubit logic gates. For example, the one or more two-qubit logic gates may include one or more two-qubit Pauli gates. In addition the one or more two-qubit logic gates, performing surface error code correction may further include implementing one or more one-qubit logic gates.

[0071] At step **706**, implementing the one or more two-qubit logic gates may include performing a plurality of measurements. Each measurement of the plurality of measurements may be a one-qubit measurement or a two-qubit measurement. Thus, each of the one or more two-qubit logic gates may be implemented without having to perform a measurement of three or more qubits. In some embodiments, for each measurement-based physical qubit included in the logical qubit encoding surface, performing the surface code error correction may include performing at most five two-qubit measurements at that physical qubit, as shown at step **708**. Additionally, for each measurement-based physical qubit included in the logical qubit encoding surface, performing the surface code error correction may include performing at most four two-qubit measurements at that physical qubit, as shown at step **710**.

[0072] The devices and methods described above allow surface codes to be used for error correction at logical qubit encoding surfaces constructed with measurement-based qubits. Previously, surface codes were only usable with CNOT-based qubits. In addition, the devices and methods described above allow surface-code-compatible logical qubit encoding surfaces to be constructed with small numbers of physical qubits and low connectivity degrees between qubits. Thus, the above devices and methods allow quantum computations to be performed with low error rates and efficient error correction.

[0073] In some embodiments, the methods and processes described herein may be tied to a computing system of one or more computing devices. In particular, such methods and processes may be implemented as a computer-application program or service, an application-programming interface (API), a library, and/or other computer-program product.

[0074] FIG. **17** schematically shows a non-limiting embodiment of a computing system **800** that can enact one or more of the methods and processes described above. Computing system **800** is shown in simplified form. Computing system **800** may be included in the quantum computing device **100** described above and illustrated in FIG. **5**. Computing system **800** may take the form of one or more personal computers, server computers, tablet computers, home-entertainment computers, network computing devices, gaming devices, mobile computing devices, mobile communication devices (e.g., smart phone), and/or other computing devices, and wearable computing devices such as smart wristwatches and head mounted augmented reality devices.

[0075] Computing system **800** includes a logic processor **802** volatile memory **804**, and a non-

volatile storage device **806**. Computing system **800** may optionally include a display subsystem **808**, input subsystem **810**, communication subsystem **812**, and/or other components not shown in FIG. 17.

[0076] Logic processor **802** includes one or more physical devices configured to execute instructions. For example, the logic processor may be configured to execute instructions that are part of one or more applications, programs, routines, libraries, objects, components, data structures, or other logical constructs. Such instructions may be implemented to perform a task, implement a data type, transform the state of one or more components, achieve a technical effect, or otherwise arrive at a desired result.

[0077] The logic processor may include one or more physical processors (hardware) configured to execute software instructions. Additionally or alternatively, the logic processor may include one or more hardware logic circuits or firmware devices configured to execute hardware-implemented logic or firmware instructions. Processors of the logic processor **802** may be single-core or multi-core, and the instructions executed thereon may be configured for sequential, parallel, and/or distributed processing. Individual components of the logic processor optionally may be distributed among two or more separate devices, which may be remotely located and/or configured for coordinated processing. Aspects of the logic processor may be virtualized and executed by remotely accessible, networked computing devices configured in a cloud-computing configuration. In such a case, these virtualized aspects are run on different physical logic processors of various different machines, it will be understood.

[0078] Non-volatile storage device **806** includes one or more physical devices configured to hold instructions executable by the logic processors to implement the methods and processes described herein. When such methods and processes are implemented, the state of non-volatile storage device **806** may be transformed—e.g., to hold different data.

[0079] Non-volatile storage device **806** may include physical devices that are removable and/or built-in. Non-volatile storage device **806** may include optical memory (e.g., CD, DVD, HD-DVD, Blu-Ray Disc, etc.), semiconductor memory (e.g., ROM, EPROM, EEPROM, FLASH memory, etc.), and/or magnetic memory (e.g., hard-disk drive, floppy-disk drive, tape drive, MRAM, etc.), or other mass storage device technology. Non-volatile storage device **806** may include nonvolatile, dynamic, static, read/write, read-only, sequential-access, location-addressable, file-addressable, and/or content-addressable devices. It will be appreciated that non-volatile storage device **806** is configured to hold instructions even when power is cut to the non-volatile storage device **806**.

[0080] Volatile memory **804** may include physical devices that include random access memory. Volatile memory **804** is typically utilized by logic processor **802** to temporarily store information during processing of software instructions. It will be appreciated that volatile memory **804** typically does not continue to store instructions when power is cut to the volatile memory **804**.

[0081] Aspects of logic processor **802**, volatile memory **804**, and non-volatile storage device **806** may be integrated together into one or more hardware-logic components. Such hardware-logic components may include field-programmable gate arrays (FPGAs), program-and application-specific integrated circuits (PASIC/ASICs), program-and application-specific standard products (PSSP/ASSPs), system-on-a-chip (SOC), and complex programmable logic devices (CPLDs), for example.

[0082] The terms “module,” “program,” and “engine” may be used to describe an aspect of computing system **800** typically implemented in software by a processor to perform a particular function using portions of volatile memory, which function involves transformative processing that specially configures the processor to perform the function. Thus, a module, program, or engine may be instantiated via logic processor **802** executing instructions held by non-volatile storage device **806**, using portions of volatile memory **804**. It will be understood that different modules, programs, and/or engines may be instantiated from the same application, service, code block, object, library, routine, API, function, etc. Likewise, the same module, program, and/or engine may be instantiated

by different applications, services, code blocks, objects, routines, APIs, functions, etc. The terms “module,” “program,” and “engine” may encompass individual or groups of executable files, data files, libraries, drivers, scripts, database records, etc.

[0083] When included, display subsystem **808** may be used to present a visual representation of data held by non-volatile storage device **806**. The visual representation may take the form of a graphical user interface (GUI). As the herein described methods and processes change the data held by the non-volatile storage device, and thus transform the state of the non-volatile storage device, the state of display subsystem **808** may likewise be transformed to visually represent changes in the underlying data. Display subsystem **808** may include one or more display devices utilizing virtually any type of technology. Such display devices may be combined with logic processor **802**, volatile memory **804**, and/or non-volatile storage device **806** in a shared enclosure, or such display devices may be peripheral display devices.

[0084] When included, input subsystem **810** may comprise or interface with one or more user-input devices such as a keyboard, mouse, touch screen, or game controller. In some embodiments, the input subsystem may comprise or interface with selected natural user input (NUI) componentry. Such componentry may be integrated or peripheral, and the transduction and/or processing of input actions may be handled on-or off-board. Example NUI componentry may include a microphone for speech and/or voice recognition; an infrared, color, stereoscopic, and/or depth camera for machine vision and/or gesture recognition; a head tracker, eye tracker, accelerometer, and/or gyroscope for motion detection and/or intent recognition; as well as electric-field sensing componentry for assessing brain activity; and/or any other suitable sensor.

[0085] When included, communication subsystem **812** may be configured to communicatively couple various computing devices described herein with each other, and with other devices. Communication subsystem **812** may include wired and/or wireless communication devices compatible with one or more different communication protocols. As non-limiting examples, the communication subsystem may be configured for communication via a wireless telephone network, or a wired or wireless local-or wide-area network, such as a HDMI over Wi-Fi connection. In some embodiments, the communication subsystem may allow computing system **800** to send and/or receive messages to and/or from other devices via a network such as the Internet.

[0086] According to one aspect of the present disclosure, a quantum computing device is provided, including a logical qubit encoding surface including a plurality of plaquettes. Each plaquette of the plurality of plaquettes may include a plurality of measurement-based qubits. The plurality of measurement-based qubits may include four data qubits and a first ancilla qubit. The first ancilla qubit may be electrically connected to the four data qubits and a second ancilla qubit included in the logical qubit encoding surface.

[0087] According to this aspect, the four data qubits of the plaquette may be arranged in a square.

[0088] According to this aspect, the first ancilla qubit may be located within the square.

[0089] According to this aspect, the respective data qubits included in the plurality of plaquettes may be arranged in a rectangular grid.

[0090] According to this aspect, for each plaquette of the plurality of plaquettes, an electrical connection between the first ancilla qubit and the second ancilla qubit may extend in a direction perpendicular to an edge of the rectangular grid that is closest to a midpoint between the first ancilla qubit and the second ancilla qubit.

[0091] According to this aspect, the quantum computing device may further include a measurement device configured to perform a measurement of a logical qubit encoded by the logical qubit encoding surface. The measurement device may be further configured to measure a respective stabilizer operator of each plaquette included in the logical qubit encoding surface. The quantum computing device may further include a classical computing device configured to identify an error in the measurement of the logical qubit based on respective measurement results of the stabilizer operators.

[0092] According to this aspect, each stabilizer operator may be a product of four Pauli X operators or four Pauli Z operators of the four respective data qubits included in the plaquette.

[0093] According to this aspect, the measurement device may be configured to measure the product of the four Pauli X operators for each of a first set of plaquettes and the product of the four Pauli Z operators for each of a second set of plaquettes. The plurality of plaquettes may be arranged in a rectangular grid including a first plurality of respective grid positions of the first set of plaquettes that alternate with a second plurality of respective grid positions of the second set of plaquettes.

[0094] According to this aspect, for each plaquette, the measurement device may be configured to measure the respective stabilizer operator of the plaquette at least in part by preparing the second ancilla qubit to have a $|+\rangle$ state. Measuring the stabilizer operator may further include applying one or more respective controlled not (CNOT) gates to the plaquette. For each CNOT gate, a data qubit of the four data qubits is a target qubit. Measuring the stabilizer operator may further include performing a second ancilla measurement at the second ancilla qubit subsequently to applying the one or more CNOT gates.

[0095] According to this aspect, the second ancilla measurement may be a measurement of a Pauli X operator or a Pauli Z operator.

[0096] According to this aspect, applying each CNOT gate may include performing a plurality of first ancilla measurements of at least the first ancilla qubit. Each first ancilla measurement may be a one-qubit measurement or a two-qubit measurement. Applying the CNOT gate may further include, for each first ancilla measurement, subsequently to that first ancilla measurement, performing a Pauli update on at least the second first qubit.

[0097] According to another aspect of the present disclosure, a method for use with a quantum computing device is provided. The method may include, at a measurement device, performing a measurement of a logical qubit encoded by a logical qubit encoding surface including a plurality of plaquettes. Each plaquette of the plurality of plaquettes may include a plurality of measurement-based qubits. The method may further include measuring a respective stabilizer operator of each plaquette included in the logical qubit encoding surface. The method may further include, at a classical computing device, identifying an error in the measurement of the logical qubit based on respective measurement results of the stabilizer operators.

[0098] According to this aspect, each plaquette may include a first ancilla qubit and may be electrically connected to a second ancilla qubit. Measuring the stabilizer operator may include preparing the second ancilla qubit to have a $|+\rangle$ state. Measuring the stabilizer operator may further include applying one or more respective controlled not (CNOT) gates to the plaquette, wherein for each CNOT gate, a data qubit of the four data qubits is a target qubit. Measuring the stabilizer operator may further include performing a second ancilla measurement at the second ancilla qubit subsequently to applying the one or more CNOT gates.

[0099] According to this aspect, the second ancilla measurement may be a measurement of a Pauli X operator or a Pauli Z operator.

[0100] According to this aspect, applying each CNOT gate may include performing a plurality of first ancilla measurements of at least the first ancilla qubit. Each first ancilla measurement may be a one-qubit measurement or a two-qubit measurement. For each first ancilla measurement, subsequently to that first ancilla measurement, applying the CNOT gate may further include performing a Pauli update on at least the first ancilla qubit.

[0101] According to this aspect, the plurality of measurement-based qubits included in each plaquette may include four data qubits and a first ancilla qubit. The first ancilla qubit may be electrically connected to the four data qubits and a second ancilla qubit included in the logical qubit encoding surface.

[0102] According to this aspect, the stabilizer operator may include a product of four Pauli X operators or four Pauli Z operators.

[0103] According to another aspect of the present disclosure, a method for use with a quantum

computing device is provided. The method may include performing a surface code error correction at a logical qubit encoding surface including a plurality of measurement-based physical qubits. Performing the surface code error correction may include implementing one or more two-qubit logic gates. Implementing each two-qubit logic gate may include performing a plurality of measurements. Each measurement may be a one-qubit measurement or a two-qubit measurement. [0104] According to this aspect, for each measurement-based physical qubit included in the logical qubit encoding surface, performing the surface code error correction may include performing at most five two-qubit measurements at that physical qubit.

[0105] According to this aspect, for each measurement-based physical qubit included in the logical qubit encoding surface, performing the surface code error correction may include performing at most four two-qubit measurements at that physical qubit.

[0106] It will be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific routines or methods described herein may represent one or more of any number of processing strategies. As such, various acts illustrated and/or described may be performed in the sequence illustrated and/or described, in other sequences, in parallel, or omitted. Likewise, the order of the above-described processes may be changed.

[0107] The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various processes, systems and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

Claims

1. A method for use with a quantum computing device, the method comprising performing a stabilizer operator measurement of a stabilizer operator at least in part by: measuring a first Pauli operator at a first ancillary qubit; performing a first Pauli update of a second Pauli operator on the first ancillary qubit, wherein the second Pauli operator differs from the first Pauli operator; applying a first two-target controlled not (CNOT) gate at the first ancillary qubit, the second ancillary qubit, a first data qubit, and a second data qubit; applying a second two-target CNOT gate at the first ancillary qubit, the second ancillary qubit, a third data qubit, and a fourth data qubit; measuring the first Pauli operator at the first ancillary qubit; and outputting a measurement result of measuring the first Pauli operator at the first ancillary qubit.
2. The method of claim 1, wherein the first two-target CNOT gate is applied at least partially in parallel with measuring the first Pauli operator and performing the first Pauli update at the first ancillary qubit.
3. The method of claim 1, wherein the first Pauli operator is measured at the first ancillary qubit at least partially in parallel with the application of the second two-target CNOT gate.
4. The method of claim 1, wherein the first two-target CNOT gate and the second two-target CNOT gate are included in a sequence of three or more two-target CNOT gates that are applied at the quantum computing device when performing the stabilizer operator measurement.
5. The method of claim 4, wherein the three or more two-target CNOT gates are each applied at the first ancillary qubit, the second ancillary qubit, and different respective data qubits.
6. The method of claim 1, wherein the first Pauli update is a Pauli measurement conditioned on whether a preceding measurement has a nontrivial measurement outcome.
7. The method of claim 1, wherein applying the first two-target CNOT gate includes, in sequence: measuring the second Pauli operator at the second ancillary qubit; performing a second Pauli update of the first Pauli operator at the second ancillary qubit; performing a first joint Pauli measurement of the first Pauli operator at the second ancillary qubit and the second data qubit;

- performing a third Pauli update of the second Pauli operator at the second ancillary qubit; performing a second joint Pauli measurement of the second Pauli operator at the first ancillary qubit and the second ancillary qubit; performing a first joint Pauli update of the first Pauli operator at the second ancillary qubit and the second data qubit; performing a third joint Pauli measurement of the first Pauli operator at the second ancillary qubit and the first data qubit; performing a second joint Pauli update of the second Pauli operator at the first ancillary qubit and the second ancillary qubit; measuring the second Pauli operator at the second ancillary qubit; and performing a third joint Pauli update of the first Pauli operator at the second ancillary qubit and the first data qubit.
- 8.** The method of claim 1, wherein: the stabilizer operator measurement is a measurement of a stabilizer operator associated with a plaquette included in the quantum computing device; and the plaquette includes the first ancillary qubit, the second ancillary qubit, the first data qubit, the second data qubit, the third data qubit, and the fourth data qubit.
- 9.** A method for use with a quantum computing device, the method comprising: performing a stabilizer operator measurement of a stabilizer operator at least in part by: measuring a first Pauli operator at a first ancillary qubit; performing a Pauli update of a second Pauli operator on the first ancillary qubit, wherein the second Pauli operator differs from the first Pauli operator; applying a two-target controlled not (CNOT) gate at the first ancillary qubit, the second ancillary qubit, a first data qubit, and a second data qubit; performing a triple Pauli operator stabilizer measurement of the first Pauli operator at the first ancillary qubit, the second ancillary qubit, a third data qubit, a fourth data qubit, and a fifth data qubit; and outputting a measurement result of the triple Pauli operator stabilizer measurement.
- 10.** The method of claim 9, wherein the two-target CNOT gate is applied at least partially in parallel with measuring the first Pauli operator and performing the first Pauli update at the first ancillary qubit.
- 11.** The method of claim 9, wherein the two-target CNOT gate is included in a sequence of two or more two-target CNOT gates that are applied at the quantum computing device when performing the stabilizer operator measurement.
- 12.** The method of claim 11, wherein the two or more two-target CNOT gates are each applied at the first ancillary qubit, the second ancillary qubit, and different respective data qubits.
- 13.** The method of claim 9, wherein the Pauli update is a Pauli measurement conditioned on whether a preceding measurement has a nontrivial measurement outcome.
- 14.** The method of claim 1, wherein: the stabilizer operator measurement is a measurement of a stabilizer operator associated with a plaquette included in the quantum computing device; and the plaquette includes the first ancillary qubit, the second ancillary qubit, the first data qubit, the second data qubit, the third data qubit, the fourth data qubit; and a fifth data qubit.
- 15.** A computing system comprising: a quantum computing device; and a classical computing device configured to control a quantum computing device to perform a stabilizer operator measurement of a stabilizer operator at least in part by: measuring a first Pauli operator at a first ancillary qubit; performing a Pauli update of a second Pauli operator on the first ancillary qubit, wherein the second Pauli operator differs from the first Pauli operator; applying a first two-target controlled not (CNOT) gate at the first ancillary qubit, the second ancillary qubit, a first data qubit, and a second data qubit; applying a second two-target CNOT gate at the first ancillary qubit, the second ancillary qubit, a third data qubit, and a fourth data qubit; measuring the first Pauli operator at the first ancillary qubit; and outputting a measurement result of measuring the first Pauli operator at the first ancillary qubit.
- 16.** The computing system of claim 15, wherein the classical computing device is configured to control the quantum computing device to apply the first two-target CNOT gate at least partially in parallel with measuring the first Pauli operator and performing the first Pauli update at the first ancillary qubit.
- 17.** The computing system of claim 15, wherein the classical computing device is configured to

control the quantum computing device to measure the first Pauli operator at the first ancillary qubit at least partially in parallel with the application of the second two-target CNOT gate.

18. The computing system of claim 15, wherein the classical computing device is configured to control the quantum computing device to apply the first two-target CNOT gate and the second two-target CNOT gate in a sequence of three or more two-target CNOT gates that are applied at the quantum computing device when performing the stabilizer operator measurement.

19. The computing system of claim 15, wherein the Pauli update is a Pauli measurement conditioned on whether a preceding measurement has a nontrivial measurement outcome.

20. The computing system of claim 15, wherein: the stabilizer operator measurement is a measurement of a stabilizer operator associated with a plaquette included in the quantum computing device; and the plaquette includes the first ancillary qubit, the second ancillary qubit, the first data qubit, the second data qubit, the third data qubit, and the fourth data qubit.
